# Merit Education Group

## The Distributive Property

Algebra 1 · Pacing ID: DistributiveProp

Standard: 9.3.5.7

Standard: 9.3.5.8

## Lesson Overview

| | |
|---|---|
| **Objectives** | • Apply the distributive property to expand and simplify algebraic expressions.<br>• Identify equivalent expressions by distributing constants across parentheses.<br>• Connect algebraic manipulation to real-world assembly and agricultural contexts. |
| **Key Vocabulary** | Distributive Property, Equivalent Expressions, Coefficient, Like Terms, Parentheses |
| **Skills Targeted** | Distribution of constants across parentheses; sign handling with negatives; simplification after distribution |
| **Materials** | Whiteboards & dry-erase markers · Student notebooks & pencils · Tiered Independent Worksheet (T1/T2/T3) · Exit Ticket slips · Color-coded distribution strips (SPED/ELL accommodation) · Projector/slides for Direct Instruction & Culture-Career context |

## Phase 1: Activate & Explore 16 min

### 🔔 Bell Ringer

4 min

See `U1L10_DistributiveProp_BellRinger.md`. Students solve on whiteboards. Collect responses to gauge readiness for algebraic distribution.

Companion: See bell ringer problem set in LMS.

### 📖 Direct Instruction

6 min

**Distributive Property:** For any real numbers $a$, $b$, $c$: $a(b + c) = ab + ac$ and $a(b - c) = ab - ac$

Emphasize that distribution is fundamentally repeated multiplication. The constant outside the parentheses must multiply every term inside.

Introduce vocabulary: coefficient (numerical factor), equivalent expressions (expressions with equal value for all variable inputs).

Model sign handling explicitly: $-3(x - 4) = (-3)(x) + (-3)(-4) = -3x + 12$. Highlight how subtracting a negative yields addition.

Precision Language: Distribution always applies to every term inside parentheses. Use the area model to illustrate: the outside constant multiplies the entire width of the grouped expression.

### 🌍 Culture & Career Spotlight

6 min

#### Ojibwe Context — Manoomin Harvesting

In Ojibwe tradition, wild rice (manoomin) is harvested, dried, and packaged for community distribution. A processor creates gift bundles containing $x$ pounds of roasted manoomin and 2 pounds of dried beans. If a family orders $n$ identical bundles, the total weight $W$ (in pounds) is modeled by: W = n(x + 2) Task: Expand this expression to show how the total weight distributes across each component per bundle.

#### Career: The Toro Co., Bloomington

Assembly Line Technicians at The Toro Co. optimize packaging efficiency for lawn equipment components. Using distribution, technicians calculate total crate weights and shipping costs before loading. HS graduates enter as Assembly Line Techs with an average starting salary of $45,870, advancing into logistics and quality assurance roles that rely on rapid algebraic estimation.

## Phase 2: Practice & Apply 26 min

### 🧭 Guided Practice

8 min

**Example 1 (Basic Expansion):**
$5(3m + 7)$
$\Rightarrow 5(3m) + 5(7) = 15m + 35$

**Example 2 (Subtraction & Negatives):**
$-4(2p - 6)$
$\Rightarrow -4(2p) + (-4)(-6) = -8p + 24$
🧮 GEMDAS Reminder: Multiplication occurs before addition/subtraction. Distribute first, then simplify signs.

**Example 3 (Simplify After Distribution):**
$3(4k + 1) - k$
$\Rightarrow 12k + 3 - k = 11k + 3$
🧮 GEMDAS Reminder: Combine like terms only after distribution is complete.

### 📝 Independent Worksheet + Circulate & Support

18 min

**Tiered Structure:**

| | |
|---|---|
| T1 Foundational | Single-term distribution, positive integers. E.g., $2(x+5)$. |
| T2 Application | Negative coefficients and subtraction. E.g., $-3(4y-2)$. |
| T3 Extension | Multi-step simplification with like terms. E.g., $5(2a+3)-a+4$. |

Teacher Action: Circulate systematically. Target students stuck on sign distribution. Use probing questions rather than direct answers. Project a silent timer during independent block.

## Phase 3: Assess & Reflect 8 min

### 🎲 Zero-Device Activity — Distributive Check

5 min

Partner "Distributive Check": Students swap worksheets, verify one T2 and one T3 problem using the distributive property rules. Partners must verbally explain each multiplication step before marking correct/incorrect.

### 🎫 Exit Ticket

2 min

Expand and simplify: $-6(3x-4)+2x$
Show all distribution steps clearly.

### 🔮 Today & Tomorrow

1 min

**Today:** We used the distributive property to expand expressions and reveal equivalent forms.

**Tomorrow:** We will reverse the process using factoring and identify common factors in polynomial expressions.

## SPED & ELL Cross-Reference

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Provide color-coded distribution strips (e.g., red for outside constant, blue for inner terms) to visually track multiplication paths. Allow graphing calculator verification for T2/T3 problems. ]

Pre-teach vocabulary with bilingual glossary cards. Use sentence frames: **“I distribute ____ to both ____ and ____. The new expression is ____.”** Pair ELL students with structured peer support during circulate phase. ]

## Misconceptions & Redirect Questions

| | Misconception | Redirect Question |
|---|---|---|
| **Corrective Strategy** | Distributing only to the first term inside parentheses | **“What happens to the second term when you multiply by the outside factor? Would skipping it change the total value?”** |
| Use area model or color-coding to show distribution across all terms. | Forgetting to distribute negative signs across subtraction | **“When −2 multiplies −5, what is the product? How does that sign interact with the operation outside?”** |
| Explicitly write −1 coefficient; practice with integer number lines. | Combining like terms before distributing | **“According to order of operations, which step must happen first: multiplication inside parentheses or combining variables?”** |
| Emphasize GEMDAS: distribute first, then combine like terms. | | |

## Materials Checklist

- ☐ Whiteboards & dry-erase markers
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- ☐ Exit Ticket slips
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